

Differentiate your Objective

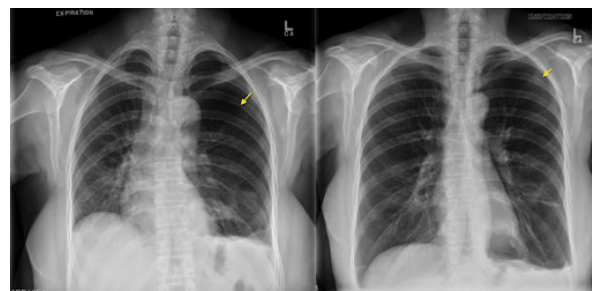
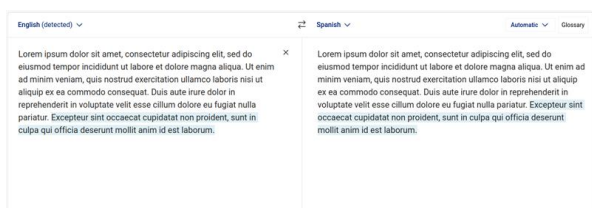
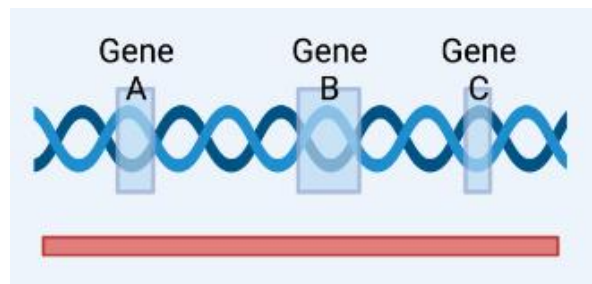
VLC = $\int \int \int$ Vision
Learning & Control

AICE3002 Introduction to Deep Learning
AICE6001 Introduction to Deep Learning (MSc)

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What does Deep Learning (and Differentiable Programming) give us?



Machine Learning in a Nutshell

All credit for this slide goes to Niranjan

Data $\{\mathbf{x}_n, \mathbf{y}_n\}_{n=1}^N \quad \{\mathbf{x}_n\}_{n=1}^N$

Function Approximator $\mathbf{y} = f(\mathbf{x}; \boldsymbol{\theta})$

Parameter Estimation $E_0 = \sum_{n=1}^N \{\|\mathbf{y}_n - f(\mathbf{x}_n; \boldsymbol{\theta})\|\}^2$

Prediction $\hat{\mathbf{y}}_{N+1} = f(\mathbf{x}_{N+1}, \hat{\boldsymbol{\theta}})$

What is Deep Learning?

Deep learning is primarily characterised by function compositions:

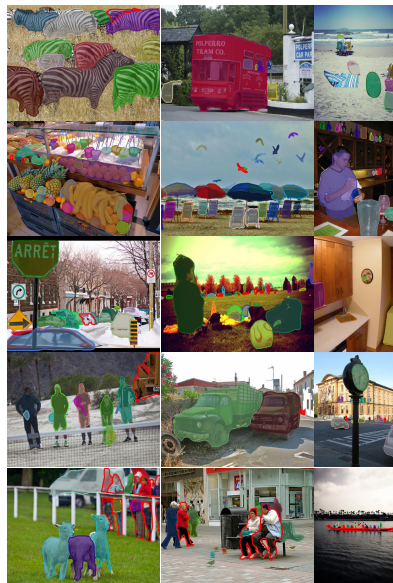
- Feedforward networks: $\mathbf{y} = f(g(\mathbf{x}; \boldsymbol{\theta}_g); \boldsymbol{\theta}_f)$
 - Often with relatively simple functions (e.g. $f(\mathbf{x}; \boldsymbol{\theta}_f) = \sigma(\mathbf{x}^\top \boldsymbol{\theta}_f)$)
- Recurrent networks:
 $\mathbf{y}_t = f(\mathbf{y}_{t-1}, \mathbf{x}_t; \boldsymbol{\theta}) = f(f(\mathbf{y}_{t-2}, \mathbf{x}_{t-1}; \boldsymbol{\theta}), \mathbf{x}_t; \boldsymbol{\theta}) = \dots$

What is Differentiable Programming?

- Captures the idea that computer programs can be partially learned rather than fully programmed by humans.
- The program is made of functions with optimisable parameters.
- So we need to be able to compute gradients with respect to the parameters of these functional blocks.
- The functional blocks don't need to be direct functions in a mathematical sense; more generally they can be *algorithms*.
- What if the functional block we're learning parameters for is itself an algorithm that optimises the parameters of an internal algorithm using a gradient based optimiser?!¹

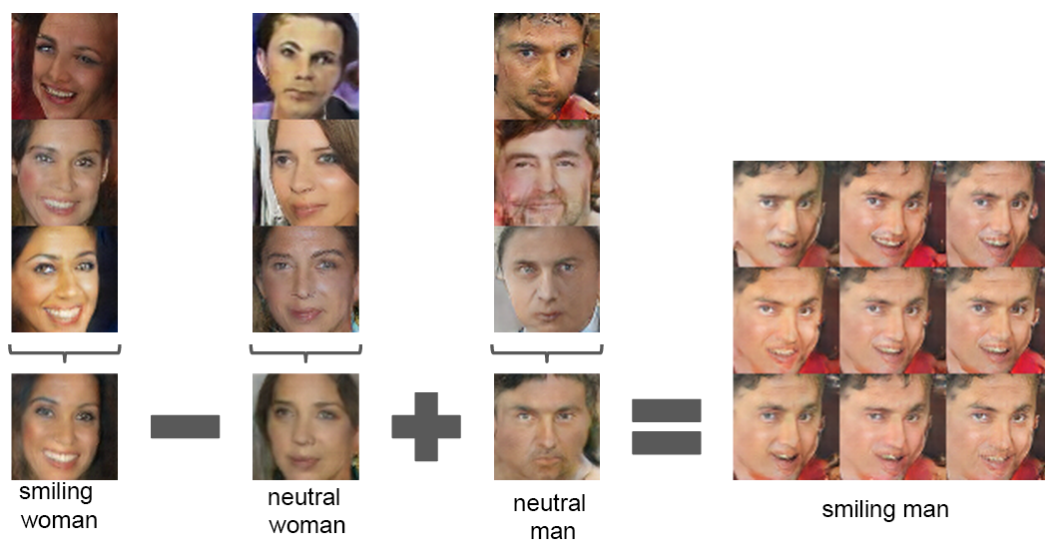
¹See our ICLR 2019 paper: <https://arxiv.org/abs/1812.03928> and NeurIPS 2019 paper: <https://arxiv.org/abs/1906.06565>

Success stories - Object detection and segmentation



Pinheiro, Pedro O., et al. "Learning to refine object segments." European Conference on Computer Vision. Springer, 2016.

Success stories - Image generation



Radford, Alec, Luke Metz, and Soumith Chintala. "Unsupervised representation learning with deep convolutional generative adversarial networks." arXiv preprint arXiv:1511.06434 (2015).

Success stories - Translation

GPT-4

Prompt:

Translate the following Irish text to English:

"Bhí mé ag siúl sa choill nuair a chuala mé ceol álainn ón adharc."

Output (With LLM Integration):

I was walking in the forest when I heard beautiful music from the horn.

Output (Traditional MT):

I was in the wood when I heard nice sound from the horn.

Lyu, C., Du, Z., Xu, J., Duan, Y., Wu, M., Lynn, T., Aji, A.F., Wong, D.F., Liu, S. and Wang, L., 2023. A Paradigm Shift: The Future of Machine Translation Lies with Large Language Models. LREC-COLING 2024

Why should we care about this?

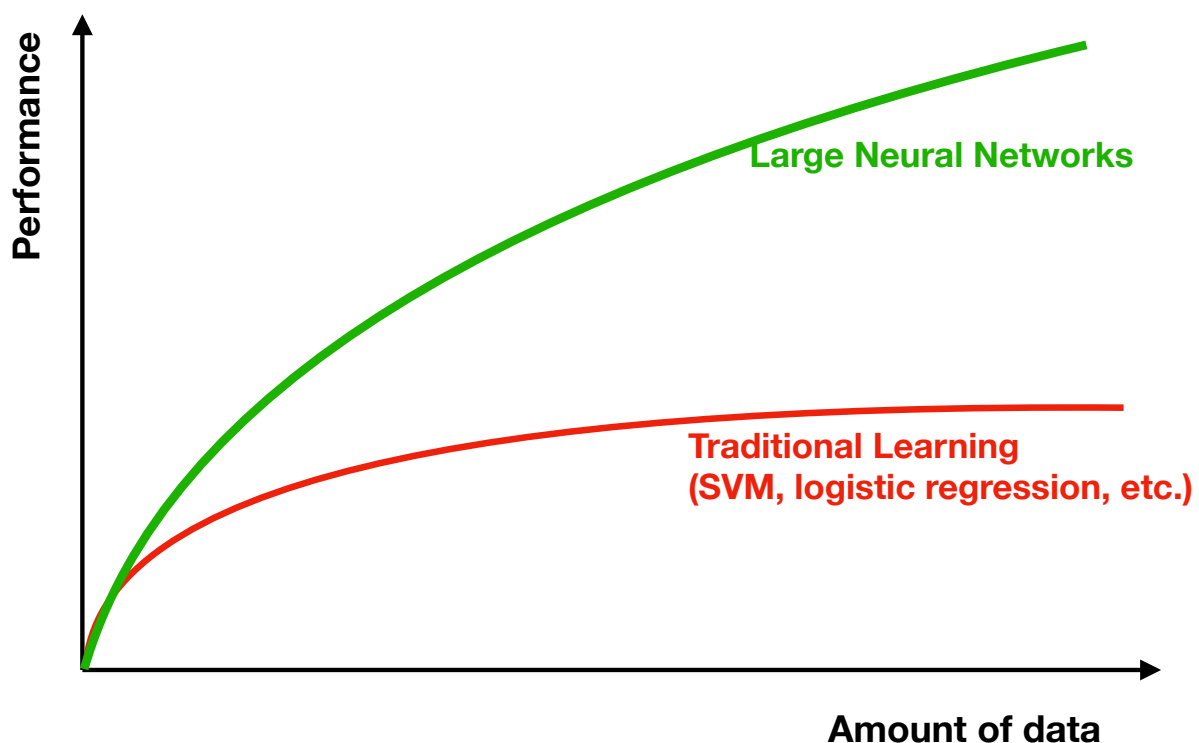


Image courtesy of Andrew Ng

Key takeaways

- **More data availability** in all domains...
- ...allows for **bigger/deeper/more complex models**
- ...models that can **learn the features**
- and these can have far **greater performance**

What is the objective of this module?

A word of warning: This is not a module about how to apply someone else's deep network architecture to a task, or how to train existing models!

You will learn some of that along the way of course, but the real objective is for you to graduate understanding how deep learning works beneath the hood.

What is the objective of this module?

- Understand the underlying mathematical and algorithmic principles of deep learning.
- Apply deep learning methods to real datasets.
- Learn how to train models using deep learning libraries.
- Think critically about why certain approaches were designed in the way they were.

How is this module going to be delivered?

- Lectures and seminars (3 most weeks)
 - This is a new module; we're utilising some material from our older module, but also adding/expanding and changing. Expect the moodle pages to change!
 - You need to read the suggested papers/links before the lectures!
 - There is maybe a little room for some flexibility later in the course on topics - tell us what you're interested in!
 - Lectures will be face to face, but also recorded for the website when possible (do not rely on this!).

How is this module going to be delivered?

- Seminars

- We'll look at a particular paper, papers or ideas in some detail and have an open discussion
- You'll need to prepare in advance & be ready to ask questions and share your thoughts
- Not recorded, but contents examinable

How is this module going to be delivered?

- Labs (1x 2 hour session per week for 8 weeks + one additional optional help / catchup session)
 - Labs consist of a number of Jupyter notebooks you will work through.
 - You'll be using PyTorch as the primary framework, with Torchbearer to help out.
 - You will need to utilise GPU-compute for the later labs (we provide Google Colab links so you can use NVidia GPUs in the cloud).
 - Labs are in-person (AICE lab in B32) with a team of PhD student demonstrators & both of us.
 - Please ask lots of questions and use this time to get help on the labs.
 - After each block of labs you will have to do a follow-up "Lab Investigation" that will be marked.

WEEK PLANNER						2026-27
Modules: AICE3002 / AICE6001						
Term Week	Academic Week	Monday	Tuesday	Wednesday	Thursday	Friday
1	1	21-Sep	22-Sep	23-Sep	24-Sep	25-Sep
			09:00 07/3027 (L/R F1) Introduction		17:00 07/3031 (L/R F2) Differentiable Programming	13:00 27/2001 (L/R 1) Vector/matrix calculus, Jacobians, JVPs, and VJP
2	2	28-Sep	29-Sep	30-Sep	1-Oct	2-Oct
		09:00 07/3027 (L/R F1) Matrix completion, the Netflix Challenge, and data anomy	Lab 1: Tensors	17:00 07/3031 (L/R F2) Computing gradients I		13:00 27/2001 (L/R 1) Computing gradients II
3	3	5-Oct	6-Oct	7-Oct	8-Oct	9-Oct
		09:00 07/3027 (L/R F1) Backpropagation	Lab 2: Automatic Differentiation	17:00 07/3031 (L/R F2) Optimisation		13:00 27/2001 (L/R 1) Seminar: Memorisation
4	4	12-Oct	13-Oct	14-Oct	15-Oct	16-Oct
		09:00 07/3027 (L/R F1) Fundamentals review	Lab 3: Optimisation	17:00 07/3031 (L/R F2) Basic ML I		13:00 27/2001 (L/R 1) Basic ML II
5	5	19-Oct	20-Oct	21-Oct	22-Oct	23-Oct
		09:00 07/3027 (L/R F1) Going deep	Lab 4: Linear Models	17:00 07/3031 (L/R F2) Practical training		lab 3&2 submission
6	6	26-Oct	27-Oct	28-Oct	29-Oct	30-Oct
		09:00 07/3027 (L/R F1) Convolution fundamentals	Lab 5: MLP	17:00 07/3031 (L/R F2) CNNs I		13:00 27/2001 (L/R 1) CNNs II
7	7	2-Nov	3-Nov	4-Nov	5-Nov	6-Nov
		09:00 07/3027 (L/R F1) CNNs III	Lab 6: CNN	17:00 07/3031 (L/R F2) Seminar: shape & texture		13:00 27/2001 (L/R 1) Embeddings
8	8	9-Nov	10-Nov	11-Nov	12-Nov	13-Nov
		09:00 07/3027 (L/R F1) Seminar: transfer learning	Lab 7: Transfer Learning	17:00 07/3031 (L/R F2) Tokenisation		13:00 27/2001 (L/R 1) RNNs
9	9	16-Nov	17-Nov	18-Nov	19-Nov	20-Nov
			XXX	Consolidation Lab Dropin	17:00 07/3031 (L/R F2) AIMA	lab 3-5 submission
10	10	23-Nov	24-Nov	25-Nov	26-Nov	27-Nov
		09:00 07/3027 (L/R F1) LSTM	Lab 8: Sequence Models	17:00 07/3031 (L/R F2) Attention		13:00 27/2001 (L/R 1) Transformers
11	11	30-Nov	1-Dec	2-Dec	3-Dec	4-Dec
		09:00 07/3027 (L/R F1) Autoencoders			XXX	XXX
12	12	7-Dec	8-Dec	9-Dec	10-Dec	11-Dec
		09:00 07/3027 (L/R F1) Revision			XXX	XXX
13, 14, 15		CHRISTMAS				
16, 17		SEMESTER 1 EXAMS				

What will we cover in the module?

AICE3002:

<https://moodle.ecs.soton.ac.uk/course/view.php?id=1031>

AICE6001:

<https://moodle.ecs.soton.ac.uk/course/view.php?id=1034>

Lab session plan

Lab	Date	Topic
Lab 1	30-Sep	Tensor programming with PyTorch
Lab 2	07-Oct	Automatic Differentiation
Lab 3	14-Oct	Optimisation
Lab 4	21-Oct	Linear Models
Lab 5	28-Oct	NNs with PyTorch and Torchbearer
	04-Nov	(Catch up / help)
Lab 6	11-Nov	CNNs with PyTorch and Torchbearer
Lab 7	18-Nov	Using pretrained models and transfer learning
Lab 8	25-Nov	Recurrent Networks, Sequence Prediction & Embeddings

Assessment Structure

- Lab Investigations 40%
- Final exam 60%

Note these are the same for both AICE3002 and AICE6001

Investigation	Submission Date	Labs covered
1 - Gradients and Matrix Completion	23rd Oct, 4PM	1 and 2
2 - Optimisation and Wide Networks	20th Nov, 4PM	3, 4 and 5
3 - Convolutions and Transfer	27th Nov, 4PM	6 and 7
4 -Transforming Sequences	11th Dec, 4PM	8

Exam

- 2 hour duration, in-person, written.
- Common Section A questions.
- Different Section B questions for AICE3002 and AICE6001.
- More details to follow later in the semester, including a cut-down sample paper with representative questions

What do we expect you already know?

- You have have taken Foundations of Machine Learning, or are currently taking it.
- Fundamentals of:
 - Matrix Algebra (matrix-matrix, matrix-vector and matrix-scalar operations, inverse, determinant, rank, Eigendecomposition, SVD);
 - Probability & Statistics (1st-order summary statistics, simple continuous and discrete probability distributions, expected values, etc); and,
 - Multivariable Calculus (partial differentiation, chain-rule).
- Programming in Python

What might you already know?

- How to use a deep learning framework (Keras, Tensorflow, PyTorch)?
- How to train an existing model architecture using a GPU?
- How to perform transfer learning?
- How to perform differentiable sampling of a Multivariate Normal Distribution?

What comes after this module?

- AICE6002: Deep Learning Research
 - taught by both of us
 - focus on translating the fundamentals you learn in this module to cutting edge research
 - plus lectures on advanced architectures, generative models, etc

Homework & questions

- Homework: Register for access to SWARM - <https://tinyurl.com/ecs-swarm>
- Any questions?